Do Now · retrieval
Hash tables and collisions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In an undirected graph, why does edge A–B appear in the adjacency list of both A and B?

Q2

In a BST, 18 is smaller than root 25. Which direction do you move?

Q3

What are valid indexes of a five-element C# array?

Q4

Calculate 17 MOD 6.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Calculate the index — 9 MOD 7 = 2, so put 9 into bucket 2.

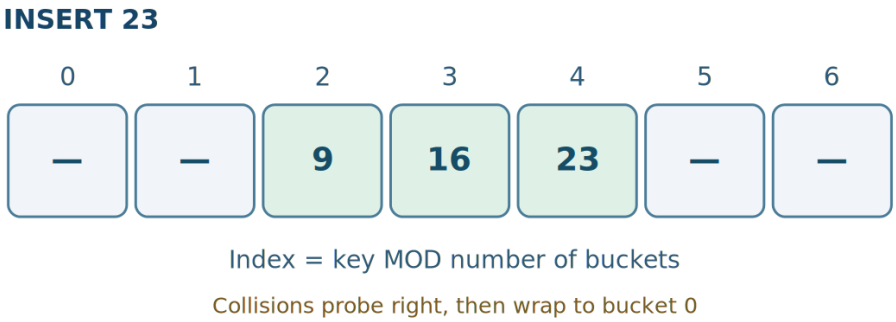
Core idea 2

Detect a collision — 16 MOD 7 is also 2. The bucket is occupied, so move to the next bucket, 3.

Core idea 3

Probe until empty — 23 also hashes to 2; buckets 2 and 3 are full, so place it in 4.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

hash function — Maps a key to an initial bucket index.

collision — Two keys map to the same initial bucket.

linear probing — Checks successive buckets, wrapping around when needed.

load factor — The proportion of occupied table buckets.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

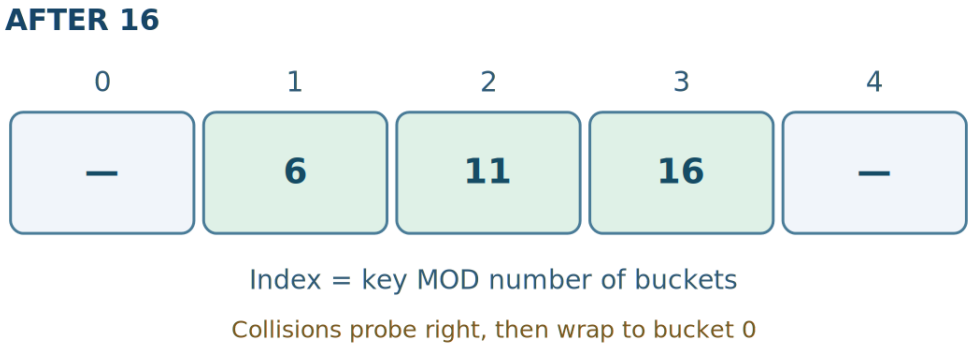
Problem

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

Follow the worked method

1. Predict and explain — Insert 6 at index 1, because 6 MOD 5 = 1.
2. Check and extend — 11 collides at 1, so store it at 2.
3. Check and extend — 16 collides at 1 and 2, so store it at 3.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict and explain
8 goes into bucket 3.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Using h(k)=k MOD 7, find the starting index for 24.

Question 2

Insert 4, 9, 14 into five empty buckets using h(k)=k MOD 5 and linear probing. Give final indexes.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain the effect of a high load factor on linear probing.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A hash collision means two different keys have the same starting bucket.

Use key vocabulary from memory

1. Explain hash function in your own words.

2. Explain collision in your own words.

3. Explain linear probing in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.